

Deliverable 8 — Adapted Architecture

Flow

Student interaction → evidence admission → hybrid learner memory → scoped retrieval → pedagogical ranking → learner-state context → response → outcome/evidence logging → maintenance.

Input layer

Capture questions, answers, corrections, goals, assessments and learning events.

Admission layer

Decide whether information is durable and record supporting evidence.

Storage layer

Use a hybrid vector + structured store with provenance.

Retrieval layer

Use semantic candidates constrained by learner, subject, skill and curriculum.

Ranking layer

Combine relevance, evidence strength, recency, uncertainty and curriculum alignment.

Context layer

Produce compact sections for strengths, misconceptions, goals and uncertainty.

Security layer

Enforce learner isolation, role permissions, auditability, deletion and retention.

Maintenance layer

Detect stale, duplicate, conflicting and low-evidence memories.

Evaluation layer

Measure retrieval, correctness, privacy, freshness and teaching usefulness.

Removed from original

The generic conversation-memory domain model is removed; education-specific learner-state entities become the core domain model.